

Stone-Weierstrass Theorem(Presentation)

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1 Introduction

The Stone Weierstrass Theorem shows that the continuous complex functions on a compact interval can be uniformly approximated by polynomials. In other words, the polynomials are uniformly dense in $C([a, b], R)$. There are several proofs which use different approaches. The most popular proof was given by Sergei Berstein, which involves the use of Bernstein polynomials. The proof we present here is very constructive and one of the most common ones.

2 Theorem: Weierstrass Approximation

Statement: $f : [a, b] \rightarrow R$ be a continuous function. Then there exists a sequence of polynomials P_n such that $P_n \rightarrow f$ uniformly on $[a, b]$.

Proof: W.L.O.G, $[a, b] = [0, 1]$. Also, we take $f(0) = f(1) = 0$, for if it is proved for this case, consider

$$g(x) = f(x) - f(0) - x[f(1) - f(0)]$$

where $g(0) = g(1) = 0$.

If g can be obtained as the limit of a sequence of polynomials, it is also true for f , since $f - g$ is a polynomial. We also take $f(x)$ to be 0 outside $[a, b]$.

Now we consider

$$Q_n(x) = c_n(1 - x^2)^n$$

where c_n are chosen such that

$$\int_{-1}^1 Q_n(x) dx = 1$$

Now,

$$\int_{-1}^1 (1-x^2)^n dx = 2 \int_0^1 (1-x^2)^n dx \geq 2 \int_0^{\frac{1}{\sqrt{(n)}}} (1-x^2)^n dx \geq 2 \int_0^{\frac{1}{\sqrt{(n)}}} (1-nx^2) dx =$$

$$\frac{4}{3\sqrt{(n)}} \geq \frac{1}{\sqrt{(n)}}$$

therefore, $c_n < \sqrt{(n)}$.

Claim: Let $\delta > 0$. Then $Q_n \rightarrow 0$ uniformly on $\delta \leq |x| \leq 1$.

Proof:

$$0 \leq 1 - x^2 \leq 1 - \delta^2 \Rightarrow Q_n(x) \leq \sqrt{n}(1 - \delta^2)^n$$

Now, $0 < 1 - \delta^2 < 1$, so the term on RHS $\rightarrow 0$ as $n \rightarrow \infty$.
Hence, proved.

We construct

$$\begin{aligned} P_n(x) &= \int_{-1}^1 f(x+t)Q_n(t)dt \\ &= \int_{-1}^{-x} f(x+t)Q_n(t)dt + \int_{-x}^{1-x} f(x+t)Q_n(t)dt + \int_{1-x}^1 f(x+t)Q_n(t)dt \end{aligned}$$

$-1 \leq t \leq -x$, thus we get $x-1 \leq t+x \leq 0$ and $1 \leq t+x \leq 1+x$

However, f vanishes on the left side and right side of 0 and 1 respectively. Thus the first and third integrals vanish.

Therefore,

$$P_n(x) = \int_{-x}^{1-x} f(x+t)Q_n(t)dt = \int_0^1 f(t)Q_n(t-x)dt$$

We now show that this $P_n \rightarrow f$ Note that f is uniformly continuous. Thus, $\forall \epsilon \geq 0, \exists \delta \geq 0$ such that $|y-x| \leq \delta \Rightarrow |f(y)-f(x)| \leq \frac{\epsilon}{2}$.

$$\begin{aligned} |P_n(x) - f(x)| &= \left| \int_{-1}^1 [f(x+t) - f(x)]Q_n(t)dt \right| \leq \int_{-1}^1 |f(x+t) - f(x)|Q_n(t)dt \\ &= \int_{-1}^{-\delta} |f(x+t) - f(x)|Q_n(t)dt + \int_{-\delta}^{+\delta} |f(x+t) - f(x)|Q_n(t)dt + \int_{\delta}^1 |f(x+t) - f(x)|Q_n(t)dt \\ &= I_1 + I_2 + I_3 \text{ (let)} \end{aligned}$$

Let $M = \sup(f)$

$$I_2 = \int_{-\delta}^{+\delta} |f(x+t) - f(x)|Q_n(t)dt = \frac{\epsilon}{2} \int_{-\delta}^{+\delta} Q_n(t)dt \leq \frac{\epsilon}{2}$$

$$I_1 = \int_{-1}^{-\delta} |f(x+t) - f(x)| Q_n(t) dt \leq 2M \int_{-1}^{-\delta} Q_n(t) dt \leq 2M\sqrt{n}(1-\delta^2)^n$$

$$I_3 = \int_{\delta}^1 |f(x+t) - f(x)| Q_n(t) dt \leq 2M\sqrt{n}(1-\delta^2)^n$$

Therefore,

$$|P_n(x) - f(x)| \leq 4M\sqrt{n}(1-\delta^2)^n + \frac{\epsilon}{2} < \epsilon$$

for large enough n .

Hence, the theorem has been proved!